

Field Service Memo

MR

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Announcement regarding the tool software for the RM log decoding for Connected Coil

1. Abstract

This document announces the release of the Service tool software
“ConnectedCoilViewer V1.0.0.1” of RM log decoding for the connected coil, and explains
how to use it.

2. Applicable system

MRT-2003, MRT-2004, MRT-1510, MRT-2020, MRT-1550, MRT-3010, MRT-3020
All have NO dependency to auxiliary model code.

3. Applicable date

The same day of this FSM issuing

4. Contents

There are some cases that connected coil(s) is/are not recognized by the system properly because coil ID becomes incorrect due to unexpected coil connection or coil corruption. In this case, analyzing the RM log which is saved in SP from RM board (below, RM Log) is useful to find the corrupted coil due to searching the corrupted coil ID. However, to analyze RM Log which has huge and user-unfriendly log, find coil ID logs from reported date, and compare all proper coil IDs with logged coil IDs, spends many hours trouble shooting. We also have some experiences that, the phenomenon of coil ID corruption was analyzed incorrectly as some boards are corrupt, and analyzing requirement via FIXS would have been required. For the sake of reduction of analyzing load and time, we have developed the tool software "ConnectedCoilViewer", which can decode RM Log generated from RM board, and displays all connected coil names which the system recognized.

This FSM announces its release and explains how to use it below.

4.1. Purpose of this tool software

- ✓ It can decode RM Log, generated to SP, or downloaded to service PC from SP.
- ✓ It can assist CE's understanding, at which port(s) and what coil(s) is/are connected, by displaying the decode result by coil name easy-to-understand.
- ✓ This tool software is just a viewer for assistance of CE's understanding. Therefore, below functionality's are not applicable.
 - Automatic judgement about ordering corrupted receiver coil or signal receiver system's service parts.
 - Automatic/Manual downloading functionality of RM Log from system or SP.
 - Other functionalities, deviating from the above purpose.

4.2. Execution environment

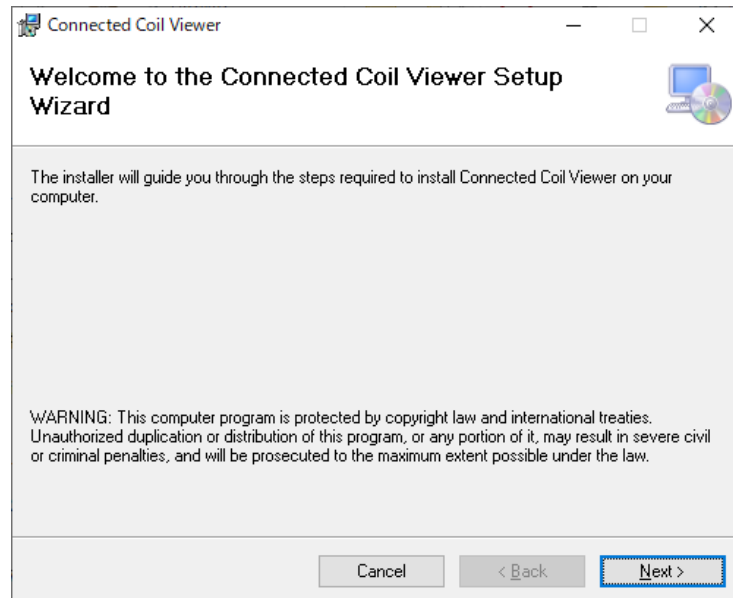
Please note to execute ConnectedCoilViewer under execution environment below. ConnectedCoilViewer has been confirmed to be able to execute and perform in service PC and SP with listed Windows and software below.

- ✓ Windows10 64bit (Japanese or English) OR
Windows7 64bit / 32bit (Japanese or English)
- ✓ The latest version's LogAnalyzer, which has installed the valid license.
- ✓ An environment where all the latest Windows Update have been applied.

4.3. Installation

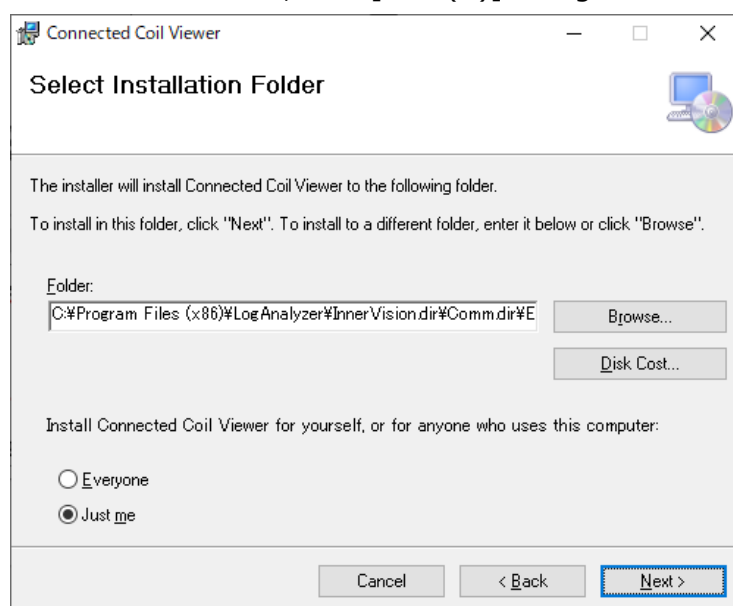
After download the Installer of ConnectedCoilViewer (below, ConnectCoilViewer.msi) from FIXS knowledge, please install following below steps.

- (1) Double-click or select ConnectCoilViewer.msi and press [Enter] key.
After showed below dialog, click [Next(N)] on right-below of dialog.



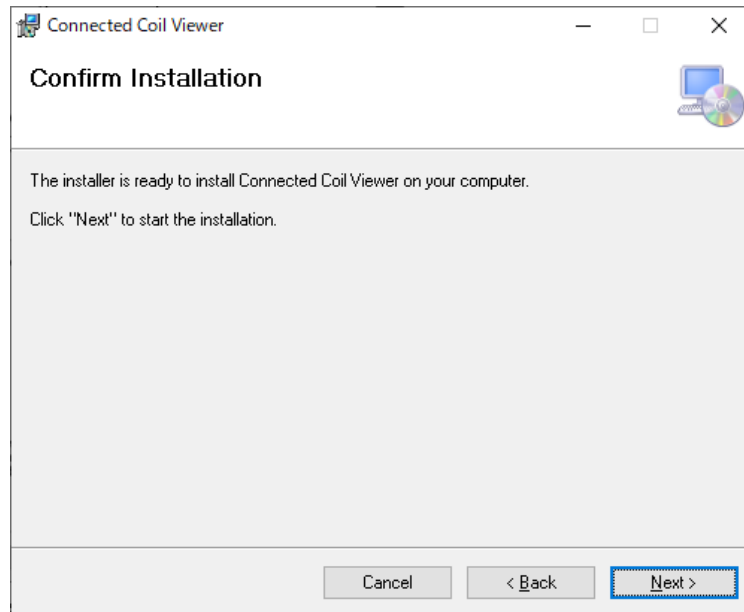
Installation entry screen of ConnectCoilViewer.msi

- (2) On "Select install folder" and selection applicable user below, **after confirmation that the same directory with installation directory of LogAnalyzer has been selected**, click [Next(N)] on right-below of dialog.



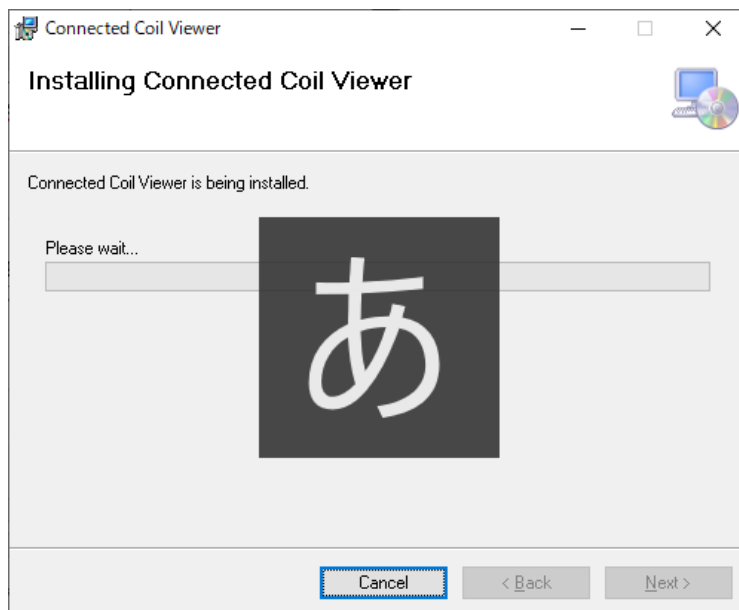
The screen of "Select install folder" and of user selection

- (3) On below screen, select [Next(N)] on right-below of dialog, start installation.



Confirmation screen before starting installation

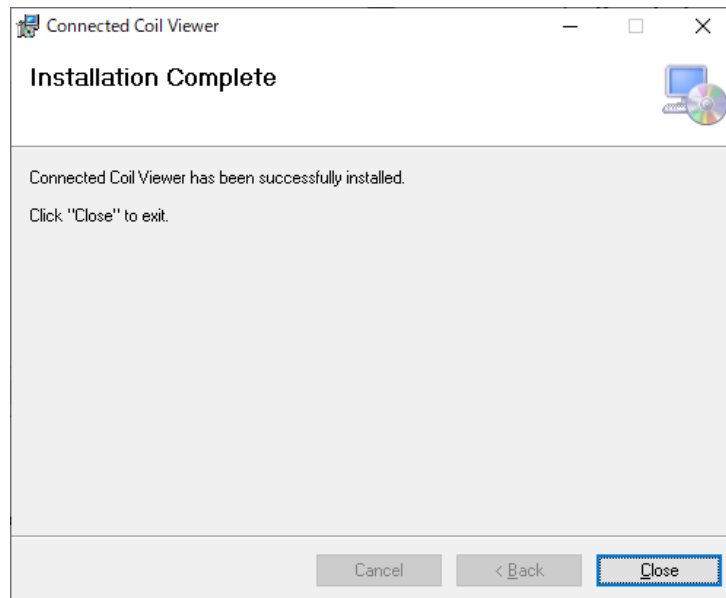
- (4) If installer starts installation, screen becomes like the example below, and progress bar under "Please wait..." message proceeds. Progress bar may be updated several times, so please wait for a while.



Installation in progress screen

- (5) If below screen is displayed, installation has been done successfully, so click [Next(N)] on right-below of dialog, and finish installation.

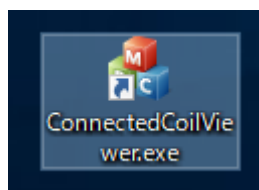
If below screen is not displayed, installation may have been failed, so restart installation steps from start (Section 4.2). If it doesn't become the screen shown below after restart installation, report phenomenon and detail steps can be found referring "5. Notes".



Installation success screen

- (6) If installation has been done successfully, short-cut icon will be generated on Desktop, invoke ConnectedCoilViewer from here.

Execution directly binary file pointed from short-cut is not recommended, because it may be affected on any other tool software which are placed on the same directory.



Short-cut on desktop

4.4. Invocation

ConnectedCoilViewer can be invoked two ways.

If the valid log file is read, discerning "the system **unequipping** coil port on couch", "the system equipping coil ports on couch", or "3T system" and receiver coil name supporting in each coil port automatically, tool software will be invoked. If tool software cannot invoke, an error description pop-up dialog may be displayed.

✓ By selecting RM Log file

If you double-click the short-cut of ConnectedCoilViewer, "Open file" dialog will be displayed. Select expected RM Log file from it, click [Open] on right-below of dialog.

For your information, this "Open file" dialog doesn't support Drag & Drop, although the short-cut icon of this tool software supports it.

✓ By Drag & Drop RM Log file

Drag & Drop the expected RM Log file on its short-cut.

CAUTION

- (1) Log file name opening by ConnectedCoilViewer should be "XXXX.RM" or "XXXX.rm".

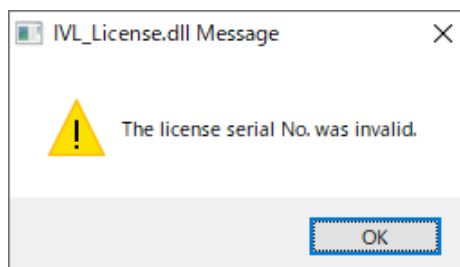
ConnectedCoilViewer cannot invoke without following this naming rule.

- (2) If any log format which ConnectedCoilViewer cannot understand is/are described in log file, the tool software cannot invoke.

- (3) If LogAnalyzer license is invalid (old, corrupted, copied, no contents, etc...), or tool software cannot evaluate LogAnalyzer license in correct (incorrect license directory, invalid license evaluation, etc.), the tool software cannot invoke.

NOTE

In case of invalidated (ex. expired) LogAnalyzer license (below, license) existence, pop-up dialog like below will be displayed. This pop-up dialog is not shown, if invalidated licenses are deleted. And tool software can invoke, if validate license exists even only one, even if this pop-up dialog is shown.

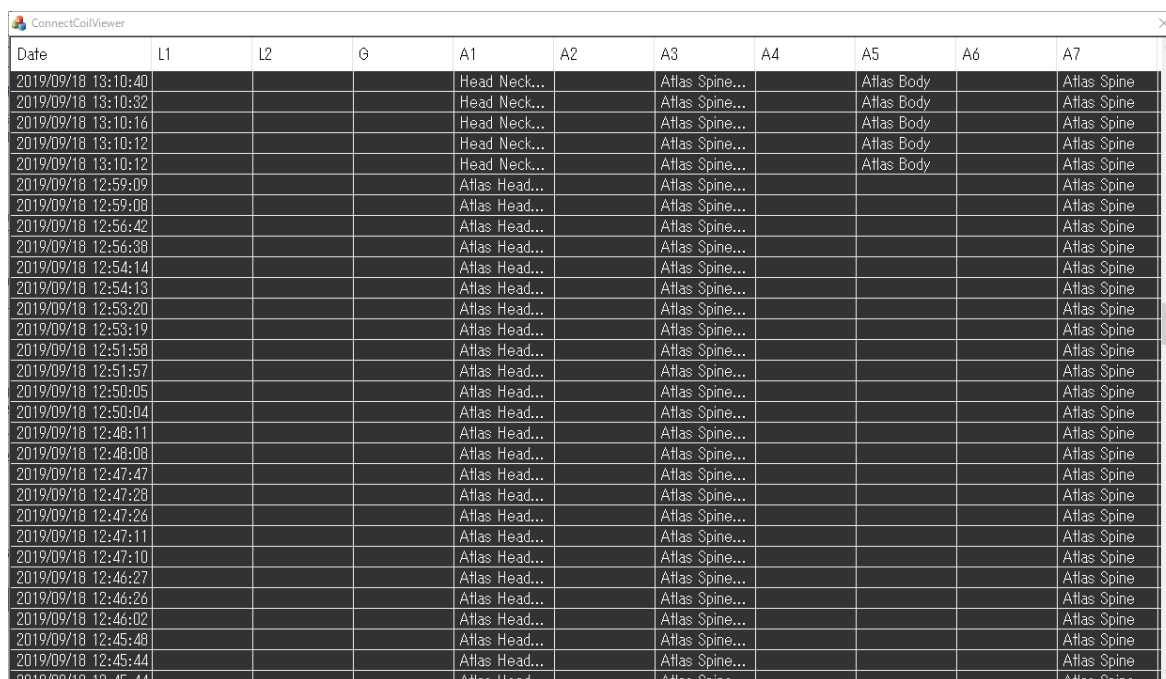


Pop-up dialog which will be displayed in case of being installed invalidated license

4.5. Running screen and operation

If ConnectedCoilViewer can be invoked successfully, the screen like bellows example will be displayed.

Here displays an example of reading RM log file generated from 3T system. The displayed contents are changed depend on MRI system type and actual connected coil(s).



Date	L1	L2	G	A1	A2	A3	A4	A5	A6	A7
2019/09/18 13:10:40				Head Neck...		Atlas Spine...		Atlas Body		Atlas Spine
2019/09/18 13:10:32				Head Neck...		Atlas Spine...		Atlas Body		Atlas Spine
2019/09/18 13:10:16				Head Neck...		Atlas Spine...		Atlas Body		Atlas Spine
2019/09/18 13:10:12				Head Neck...		Atlas Spine...		Atlas Body		Atlas Spine
2019/09/18 13:10:12				Head Neck...		Atlas Spine...		Atlas Body		Atlas Spine
2019/09/18 12:59:09				Atlas Head...		Atlas Spine...				Atlas Spine
2019/09/18 12:59:08				Atlas Head...		Atlas Spine...				Atlas Spine
2019/09/18 12:56:42				Atlas Head...		Atlas Spine...				Atlas Spine
2019/09/18 12:56:38				Atlas Head...		Atlas Spine...				Atlas Spine
2019/09/18 12:54:14				Atlas Head...		Atlas Spine...				Atlas Spine
2019/09/18 12:54:13				Atlas Head...		Atlas Spine...				Atlas Spine
2019/09/18 12:53:20				Atlas Head...		Atlas Spine...				Atlas Spine
2019/09/18 12:53:19				Atlas Head...		Atlas Spine...				Atlas Spine
2019/09/18 12:51:58				Atlas Head...		Atlas Spine...				Atlas Spine
2019/09/18 12:51:57				Atlas Head...		Atlas Spine...				Atlas Spine
2019/09/18 12:50:05				Atlas Head...		Atlas Spine...				Atlas Spine
2019/09/18 12:50:04				Atlas Head...		Atlas Spine...				Atlas Spine
2019/09/18 12:48:11				Atlas Head...		Atlas Spine...				Atlas Spine
2019/09/18 12:48:08				Atlas Head...		Atlas Spine...				Atlas Spine
2019/09/18 12:47:47				Atlas Head...		Atlas Spine...				Atlas Spine
2019/09/18 12:47:28				Atlas Head...		Atlas Spine...				Atlas Spine
2019/09/18 12:47:26				Atlas Head...		Atlas Spine...				Atlas Spine
2019/09/18 12:47:11				Atlas Head...		Atlas Spine...				Atlas Spine
2019/09/18 12:47:10				Atlas Head...		Atlas Spine...				Atlas Spine
2019/09/18 12:46:27				Atlas Head...		Atlas Spine...				Atlas Spine
2019/09/18 12:46:26				Atlas Head...		Atlas Spine...				Atlas Spine
2019/09/18 12:46:02				Atlas Head...		Atlas Spine...				Atlas Spine
2019/09/18 12:45:48				Atlas Head...		Atlas Spine...				Atlas Spine
2019/09/18 12:45:44				Atlas Head...		Atlas Spine...				Atlas Spine
2019/09/18 12:45:44				Atlas Head...		Atlas Spine...				Atlas Spine

A case example of reading and invoking RM Log file generated from 3T system

This list displays each receiver coil name connected at each coil ports, in descending order from the latest (the newest day & time's) data, by date. It displays, date, L1 port, L2 port, Gantry port, Couch A1 – A7 ports, from left of list. Single line selection is applicable to use for confirming data landmark. Selected line is displayed blue back color as below.



2019/09/18 14:30:17				Head Neck...		Atlas Spine...		Atlas Body		Atlas Spine
2019/09/18 14:30:16				Head Neck...		Atlas Spine...		Atlas Body		Atlas Spine
2019/09/18 14:28:45				Head Neck...		Atlas Spine...		Atlas Body		Atlas Spine
2019/09/18 14:28:41				Head Neck...		Atlas Spine...		Atlas Body		Atlas Spine

In case selected any a line

✓ Column width adjustment

If displayed receiver coil name is wider than cell width, “...” strings meaning not to be able to be displayed, as below left figure, is displayed. To display all strings, please try to widen the border of Header (the white top row which displays each coil port name) of expecting column till width as you want, by Drag & Drop. Double click at the border expands the width of the column till the longest strings in the column can be displayed.

A1	A1
Head Neck...	Head Neck Base
Head Neck...	Head Neck Base

In case, strings are longer than column width (left), and after widen cell width till all character is displayed (right).

✓ Display if coil ID is corrupted

If receiver coil ID become corrupted value which is not declared in coil DB of system, “Unknown” is displayed as below figure.

Head Neck...	Unknown	Atlas Spine...
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Example of display toward corrupted coil ID which isn’t declared in coil DB

On the other hand, if receiver coil ID become corrupted value which is declared in coil DB of system, the corresponding coil name toward the coil ID is displayed. For example, if 1.5T system recognized Atlas Spine coil’s coil ID (50(x16)) as Atlas Head coil’s coil ID (47(x16)) incorrectly, coil name is displayed as “Atlas Head” as below figure. Nevertheless, Atlas Head coil can connect only on A1 port, in this example, tool software can show like Atlas Head coil was connected at any coil ports not only A1 port of couch. This may happen even in case not only the port was not allowed to connect, but also Atlas Head coil was already connected A1 port of couch. Please take care.

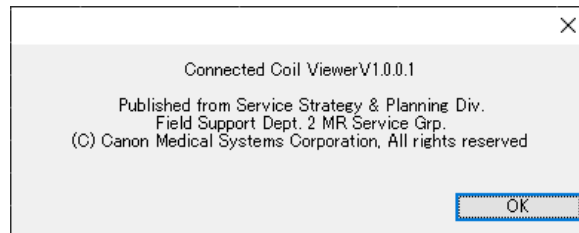
A1	A2	A3	A4
Atlas Head		Atlas Head	Atlas SpineLower
Atlas Head		Atlas Spine	Atlas SpineLower

Example that, coil ID of Atlas Spine coil which was already connected at A3 port of couch, was recognized incorrectly as Atlas Head coil’s coil ID

✓ Version information

Software version and copyright notice of invoking ConnectedCoilViewer is displayed in pop-up dialog. This pop-up dialog can be shown selecting "Version information Connected Coil Viewer (&A)..." on displayed tooltip by clicking icon at left of title bar.

Clicking "X" button at right-top of pop-up dialog or "OK" button at right-bottom of it can close this pop-up dialog.



Version information

4.6. Close

Clicking "X" button on the right-top of window can close ConnectedCoilViewer.

Typing "Alt + F4" is also available to close.

4.7. Uninstall

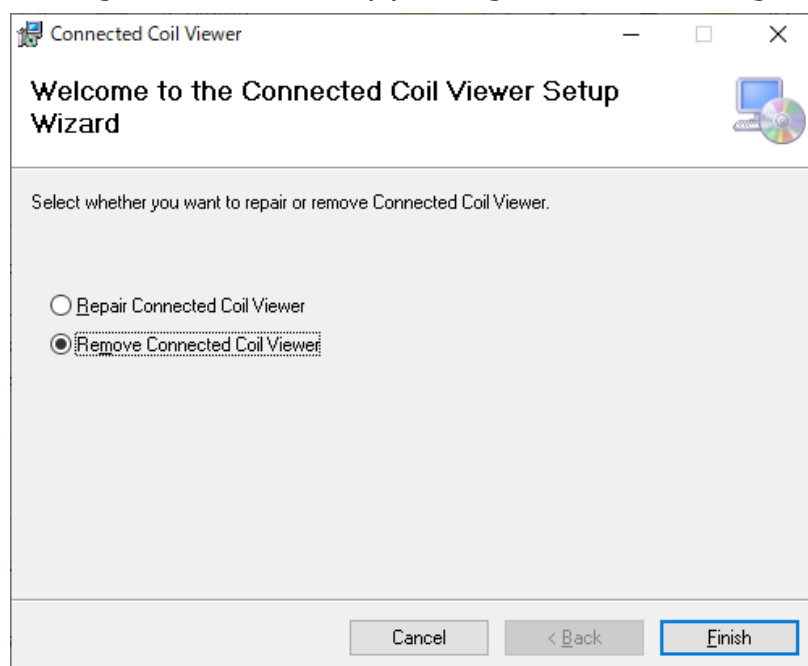
ConnectedCoilViewer can be uninstalled two ways.

- ✓ ConnectCoilViewer.msi
- ✓ Windows application management

Below explain steps for each way.

✓ Uninstall from ConnectCoilViewer.msi

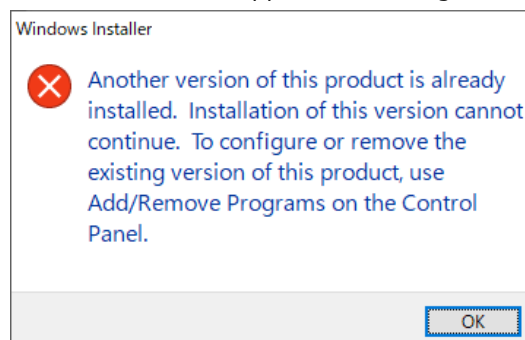
- (1) Execute ConnectedCoilViewer.msi on PC which ConnectedCoilViewer has already been installed, after select "Remove ConnectedCoilViewer (M)" on displayed dialog, and click "Finish (F)" on right-bottom of dialog.



Uninstallation entry screen of ConnectedCoilViewer

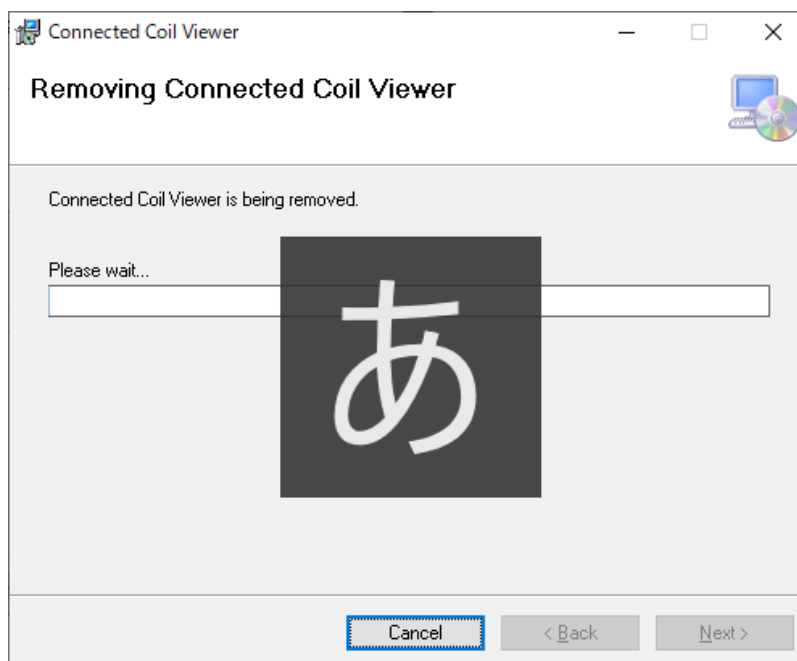
NOTE

Below dialog may be displayed, if the version between **ConnectCoilViewer.msi** and installed **ConnectedCoilViewer** is different. In this case, uninstall it following steps explained "Uninstall from Windows application management".



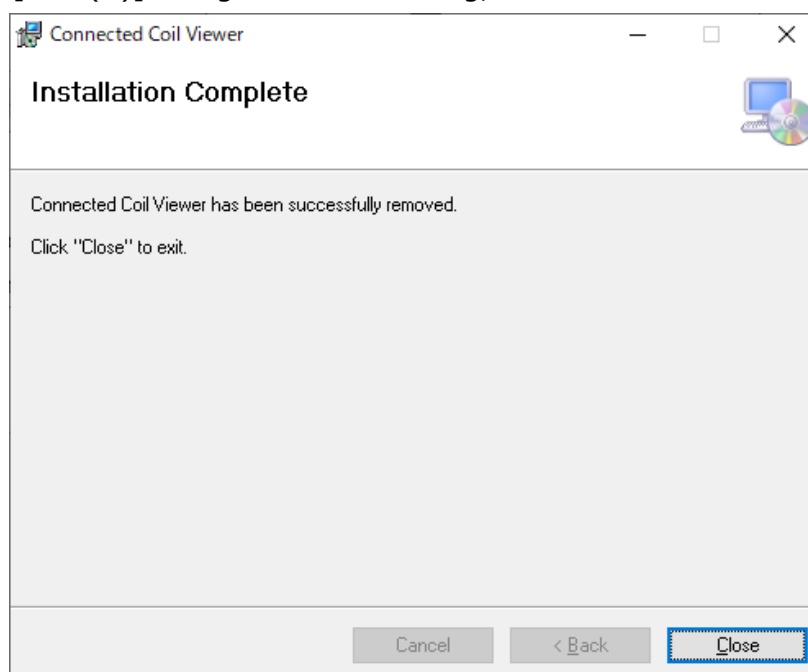
An example in case of unavailable uninstall from **ConnectCoilViewer.msi**

- (2) After starting uninstallation, below screen is displayed, and progress bar under “Please wait…” message proceeds. Progress bar may be updated in several times, so please wait for a while.



Uninstallation in progress screen

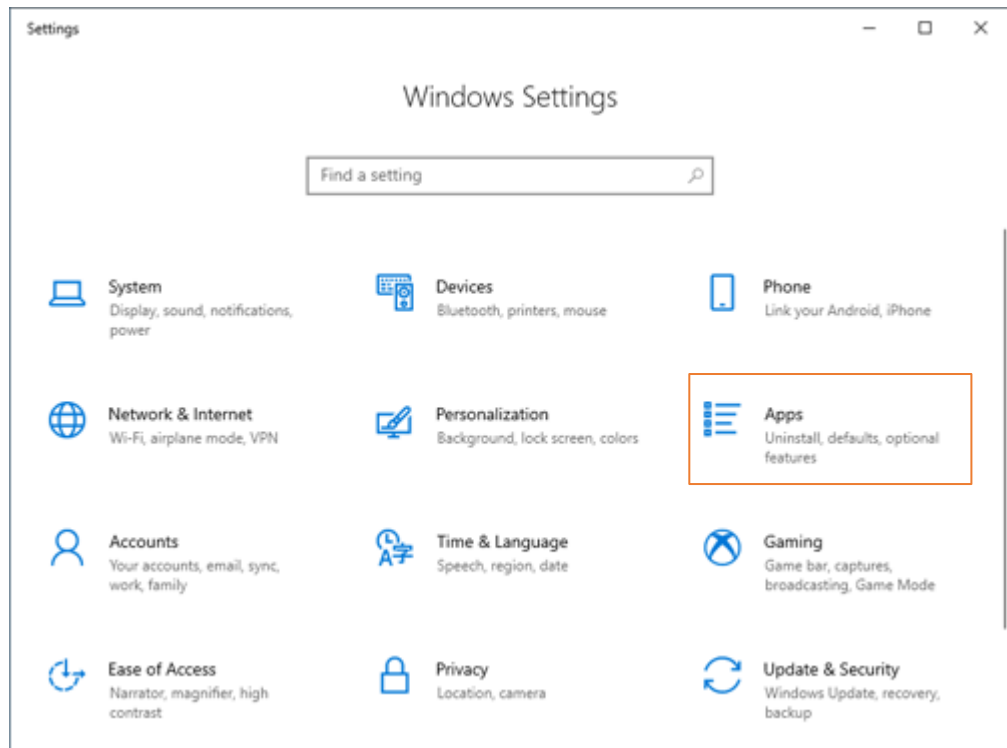
- (3) If below screen is displayed, uninstallation has been done successfully, so click [Next(N)] on right-below of dialog, and finish uninstallation.



Uninstallation success screen

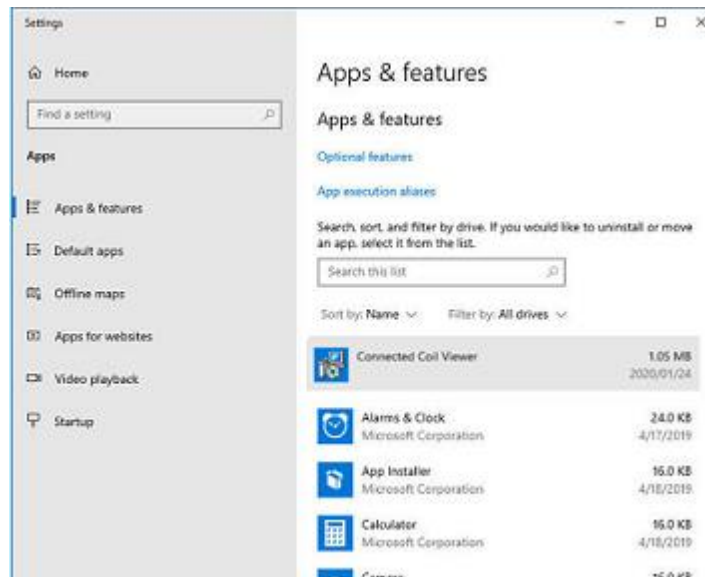
✓ **Uninstall from Windows application management**

(1) Open "Windows settings" dialog from windows start menu etc., select "Apps".



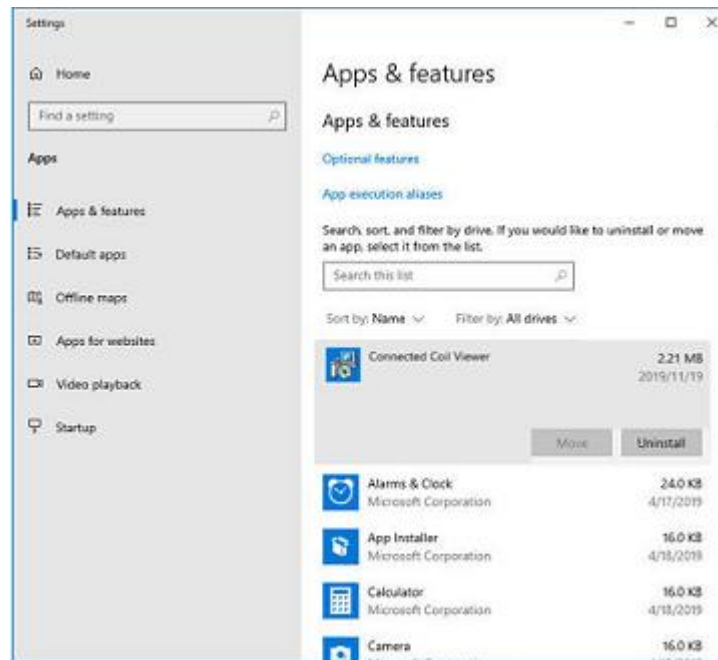
"Windows settings" screen

(2) Find and click "ConnectedCoilViewer" from installed application list bottom of "Apps & features" screen.



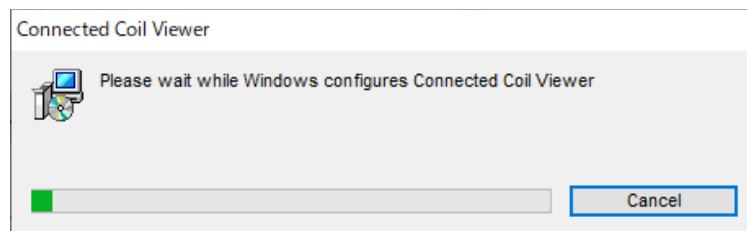
ConnectedCoilViewer which is displayed in installed application list

(3) Clicking "Uninstall" inside the selected frame, a pop-up dialog is displayed, so click "Uninstall" inside the pop-up dialog.



Screen selecting “Uninstall” on ConnectedCoilViewer

- (4) Just after displayed pop-up dialog below, uninstall will be started, so please wait for a while. If it is completed, this pop-up dialog will disappear automatically.



Pop-up dialog displayed during execution uninstall

5. Notes

- ✓ Operation, display, and design of ConnectedCoilViewer may change in the future without any announcement. However, an announcement will be made in cases of major important changes.
- ✓ If In the case of a defect report or a request for improvement for is required, please contact CMSC MR Service Group.